

Prompt Coach

Direction & Build Roadmap — planning notes for continued development

Purpose of this document: a working spec to hand to Claude Code (or reference directly) when continuing development on the Prompt Coach repository. It captures the product direction, the reasoning behind it, and a concrete build sequence — not a finished design doc.

Core Motto

“Your AI is only as smart as your prompt.”

AI output quality scales with the skill of the person prompting it, not just the model. Prompt Coach exists to close that skill gap — starting at the individual level, with an eye toward an eventual team/enterprise layer (a manager dashboard) later, once the individual product is validated.

Positioning

General-purpose, not STEM-specific. Research conversations with students (notably biology, physics, and math majors) surfaced that many don't see AI's value — not because AI isn't useful to them, but because generic advice doesn't map onto their actual work (labs, problem sets, data analysis, write-ups). Keeping the product and messaging general keeps it broadly marketable now; domain-specific framing can be layered on later without changing the core mechanic.

Sequencing: individual product and dashboard first (smaller lift, directly marketable, portfolio-ready). Enterprise/manager dashboard is a future layer — same underlying data model, aggregated across users.

Current Mechanic (already built)

A Grammarly-style Chrome extension that connects to an AI API. The user writes a prompt; the system critiques it and rewrites it before it's sent, so the response the user gets is closer to what they actually wanted than what their first draft would have produced. Currently triggered manually (an “Improve with AI” button).

Product Philosophy

- **Teach, don't just fix.** The goal isn't to repair one prompt — it's to build the user's underlying prompting ability over time, so they eventually write strong prompts on their own without the tool.
- **Micro-learning, not a course.** Bite-sized, in-the-flow coaching beats a separate lesson or curriculum. This fits a browser extension's natural form factor.
- **Ambient, not intrusive.** A “silent coach whispering in your ear,” not a blocking modal. Coaching should never interrupt the user's actual work.

- **Minimal aesthetic.** Clean, unobtrusive UI; celebrations are small (e.g. a brief confetti pop), never loud.

Key Interaction Shift: Manual → Live

Move from a manual “click to improve” button to live, ambient analysis as the user types (Grammarly-style underlining/suggestions rather than a separate action).

- **Debounce, don't stream on every keystroke.** Wait for a short pause in typing (roughly 0.5–1s) before analyzing, to avoid wasted calls and jitter.
- **Keep it snappy.** Use a fast/lightweight model for the critique step specifically (it's a much lighter task than actually answering the prompt) so live feedback doesn't feel laggy.
- **Never block the send.** The user should never feel like they're waiting on the coach before they can submit their real prompt.

Task Classifier (new)

Before coaching, classify the prompt by task type. This determines whether refinement-style coaching applies at all.

- **Generative / analytical tasks** (writing, document generation, data analysis, code, open-ended problems) → coaching and refinement nudges apply.
- **Discrete factual queries** (e.g. a single correct math answer) → no refinement follow-up needed; stay quiet.

Prompt Weakness Categories

Used both to drive real-time coaching suggestions and to build the user's long-term skill profile.

Missing context	Prompt lacks necessary data, audience, or goal information.
Vague / ambiguous ask	The request itself isn't clearly specified.
No output format specified	User doesn't state the format they want back (list, table, length, etc.).
Too broad in scope	Prompt tries to cover too much at once.
Missing constraints	No length, tone, or other bounding constraints given.

Doesn't refine / stops after one attempt

User treats the first AI output as final rather than iterating on it — e.g. gets a mediocre cover letter, concludes “AI can't do this,” and abandons it instead of steering the AI across multiple rounds (as one would when teaching AI a personal profile/preferences and iterating toward a finished document). Considered the highest-impact category — less about prompt syntax, more a mindset shift that shows up across every discipline (cover letters, lab reports, problem sets, code).

Refinement Nudge Logic

For prompts classified as generative/analytical, after the AI's first response the coach may suggest a specific refinement (e.g. “want to refine this further? Try telling it exactly what to change.”). Not triggered on every interaction — only when the task type warrants iteration. Never triggered on discrete factual answers.

User Profile & Individual Dashboard (new)

Currently the extension is fully stateless. Add lightweight local storage to build a per-user profile over time.

What to log per prompt:

- The weakness categor(ies) detected (from the list above).
- Whether the user accepted or ignored the suggestion.

Dashboard: two lenses

- **Skill snapshot** — current strengths and weaknesses by category (e.g. “strong at iterative thinking, weaker at specifying constraints”).
- **Timeline / trajectory** — per-category trend: improving, plateauing, or slipping, compared against the user's own past performance (not an external benchmark).

Feedback style

- **Improvement** → small, inline celebration (e.g. a brief confetti pop / toast). Gamified, but minimal — never a big interruptive screen.
- **Regression** → reported passively on the dashboard only. No proactive/interruptive flagging for now.

This individual dashboard is explicitly a stepping stone toward the eventual manager/enterprise dashboard (same category-based data model, later aggregated across a team) — but that is out of scope for now.

Aesthetic Direction

- Minimal overall: clean, neutral, unobtrusive — fits the “ambient coach” philosophy.
- Base palette: grays and whites.

- Accent color: cobalt blue (*placeholder — open to change later*).
- Celebration animations: small only (e.g. a brief confetti pop), never large or blocking.

Build Sequence

Ordered so each layer depends on the one before it. Goal: reach a first showable version to get outside feedback on — not a finished product.

1. Live interaction model	Replace the manual "Improve with AI" button with debounced, ambient live analysis as the user types. Use a fast/lightweight model for the critique step to keep it snappy.
2. Task classifier	Detect generative/analytical vs. discrete factual prompts before deciding whether to coach at all.
3. Coaching brain	Implement the six weakness categories and the refinement-nudge logic on top of the classifier.
4. Storage + individual dashboard	Add lightweight local storage for per-prompt weakness/acceptance data; build the skill-snapshot + timeline dashboard with minimal inline celebration for improvement.

Open / Deferred (not yet decided)

- Exact accent color beyond the cobalt-blue placeholder.
- Manager/enterprise dashboard and cross-user telemetry — deliberately out of scope until the individual product is validated.
- Whether clarifying-question flows (the coach asking the user something before rewriting) get built, and if so, how that's handled given the extension runs inside a chatbot's own input box.

Prepared from a planning conversation, for use as a working spec when continuing development (including handoff to Claude Code on the project repository).